SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	30 % of CIA	**Total Marks:**	20 (Answer ANY ONE)
CO3 Statement:	*Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)*		
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Section / Batch:	CSE(AI)	**Date:**	24-10-26

Code of Conduct

I N.V. Sandeep Kumar (192372026) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.V. Sandeep Kumar

Instructions

- Duration: 10 minutes

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1. Derive the capacity gain achieved from 3-sector sectorization, assuming interference is split equally across sectors, starting from a base case of 45 users per omnidirectional cell. Show the full calculation for the sectorized capacity. Discuss why this capacity gain in CDMA works somewhat differently from sectorization gain in FDMA/TDMA, given CDMA's interference-limited (rather than channel-limited) nature.

A. **1. Introduction**

Code Division Multiple Access (CDMA) is a mobile communication technology in which multiple users share the same frequency band using different spreading codes. Its capacity is mainly limited by interference from other users.

Sectorization is a technique that divides a cell into smaller sectors using directional antennas. In 3-sector sectorization, the cell is divided into three sectors, each covering approximately 120 degrees.

The objective of this problem is to calculate the capacity gain achieved through 3-sector sectorization, starting with an omnidirectional cell capacity of 45 users, and explain why sectorization affects CDMA differently from FDMA and TDMA.

2. Given Parameters

Parameter	Symbol	Value
Original cell capacity	(N_0)	45 users
Number of sectors	(S)	3
Interference fraction per sector	(I_s/I_0)	(1/3)
Sector coverage angle	(\theta)	(120^\circ)

3. Calculation of Sectorized Capacity

Step 1: Understand interference reduction

In an omnidirectional cell, the antenna receives signals from all directions. In a sectorized cell, directional antennas mainly receive signals from their respective coverage areas.

Assuming interference is reduced to one-third, the interference fraction is:

[
\frac{I_s}{I_0}=\frac{1}{3}
]

Step 2: Calculate the capacity of a sector

Under the simplified assumption that capacity is inversely proportional to interference:

[
N_s=\frac{N_0}{I_s/I_0}
]

Substituting the values:

[
N_s=\frac{45}{1/3}
]

[
N_s=45\times3
]

[
\boxed{N_s=135\text{users}}
]

This is the idealized capacity obtained from the assumed threefold reduction in interference. It should not be multiplied by three again unless the original 45-user figure is explicitly defined as the capacity of each sector rather than the entire omnidirectional cell.

Step 3: Calculate the capacity gain

The capacity gain factor is:

[
G=\frac{N_s}{N_0}
]

[
G=\frac{135}{45}=3
]

Therefore,

[
\boxed{G=3\text{ times}}
]

Step 4: Calculate the percentage increase

$$
\text{Percentageincrease}
=\frac{N_s-N_0}{N_0}\times100
$$

$$
=\frac{135-45}{45}\times100
$$

$$
=\frac{90}{45}\times100
$$

$$
\boxed{\text{Percentageincrease}=200\%}
$$

Interpretation: Under the simplified interference model, capacity increases from 45 to 135 users, which is a threefold capacity factor or a 200% increase.

4. Working Principle of 3-Sector Sectorization

The three-sector arrangement uses three directional antennas to cover the complete cell area.

1. The first antenna covers a sector of approximately 120 degrees.
2. The second antenna covers another 120-degree sector.
3. The third antenna covers the remaining 120-degree sector.
4. Directional antennas reduce the reception of unwanted signals from other directions.
5. Reduced interference improves the signal-to-interference ratio (SIR), allowing more simultaneous users to be supported.

The actual capacity improvement depends on antenna directionality, propagation conditions, power control, and interference from neighboring cells.

5. Difference Between CDMA and FDMA/TDMA Sectorization

Feature	**CDMA**	**FDMA/TDMA**
Capacity limitation	Mainly interference-limited	Mainly limited by available frequency channels or time slots

User separation	Spreading codes	Frequencies or time slots
Role of sectorization	Reduces interference and improves SIR	Enables better frequency or time-slot reuse
Capacity improvement	Depends on interference reduction	Depends on channel allocation and reuse planning

CDMA

In CDMA, multiple users transmit simultaneously over the same frequency band using different spreading codes. Each active user contributes interference to other users.

Sectorization reduces unwanted signals received from different directions. This improves the signal-to-interference ratio and can increase the number of simultaneous users.

Therefore, CDMA capacity depends strongly on the total interference level.

FDMA and TDMA

In FDMA, each user is assigned a separate frequency channel. In TDMA, users share a frequency channel but transmit in different time slots.

Sectorization allows channels or time slots to be reused in different directions, subject to interference constraints. However, capacity is also limited by the number of available channels or time slots.

Thus, sectorization in CDMA primarily improves interference conditions, whereas in FDMA/TDMA it helps improve the reuse of limited radio resources.

6. Practical Engineering Considerations

The calculated capacity of 135 users is an idealized result based on the assumption that interference is reduced to one-third and that capacity varies inversely with interference.

In real CDMA networks, the actual capacity gain may be lower because of:

- Imperfect directional antenna patterns.
- Interference from neighboring cells.
- User distribution and mobility.
- Power-control limitations.
- Background noise and propagation conditions.

Consequently, the calculated value should be treated as a theoretical estimate rather than a guaranteed practical capacity.

7. Final Results

Quantity	Calculated result
Original omnidirectional capacity	45 users
Assumed interference fraction	(1/3)
Idealized sectorized capacity	135 users
Capacity gain factor	3 times
Percentage increase	200%

8. Conclusion

Three-sector sectorization divides a cellular area into three 120-degree sectors using directional antennas. By reducing unwanted interference, sectorization can improve the signal-to-interference ratio and increase CDMA capacity.

Starting with an omnidirectional capacity of 45 users, the simplified model gives a sectorized capacity of 135 users, representing a threefold capacity factor and a 200% increase.

Unlike FDMA and TDMA, where capacity depends strongly on the availability and reuse of frequency channels or time slots, CDMA capacity is mainly interference-limited. Therefore, the practical gain from sectorization depends on the actual reduction in interference and the radio conditions of the network.